**EXP NO: 20** **Experiment to understand K-means clustering algorithm for a given data set. give aim for this**

**AIM:**

To understand and implement the K-means clustering algorithm on a given dataset to group similar data points into clusters based on their features.

**PROGRAM:**

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.cluster import KMeans

from google.colab import files


uploaded = files.upload()

df = pd.read_csv('Mall_Customers.csv')


df.info()

print(df.head())


sns.pairplot(df)

plt.show()


features = df.iloc[:, [3, 4]].values


model = KMeans(n_clusters=5, random_state=42, n_init=10)

model.fit(features)


Final = df.iloc[:, [3, 4]].copy()
```

```
Final['label'] = model.predict(features)


print(Final.head())


sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()


plt.show()


features_el = df.iloc[:, [2, 3, 4]].values
wcss = []


for i in range(1, 10):
    model = KMeans(n_clusters=i, random_state=42, n_init=10)
    model.fit(features_el)
    wcss.append(model.inertia_)


plt.plot(range(1, 10), wcss, marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method')
plt.show()
```

**OUTPUT:**

```
#   Column                Non-Null Count  Dtype
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```

**0 CustomerID 200 non-null int64**

**1 Gender 200 non-null object**

**2 Age 200 non-null int64**

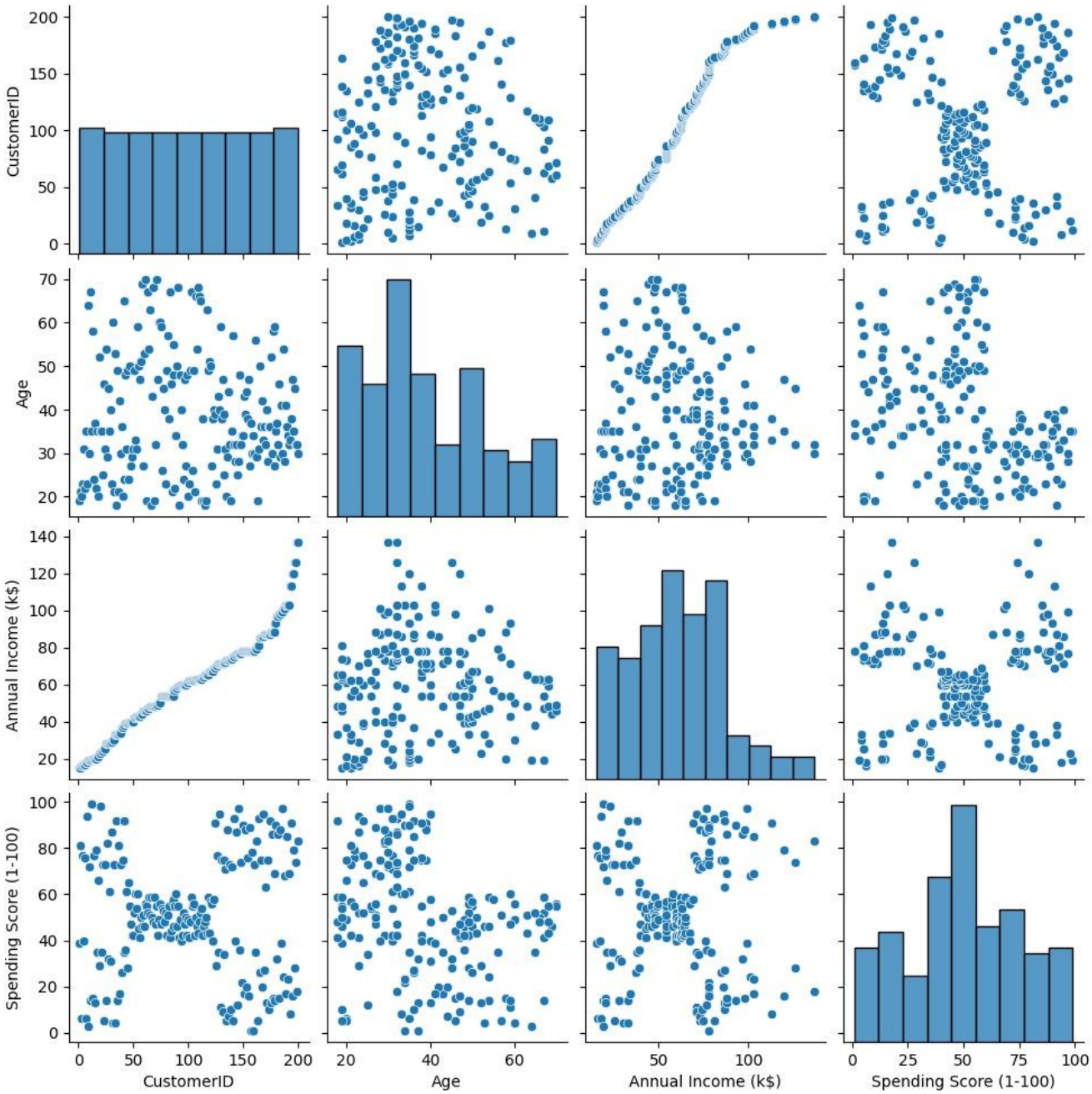
**3 Annual Income (k$) 200 non-null int64**

**4 Spending Score (1-100) 200 non-null int64**

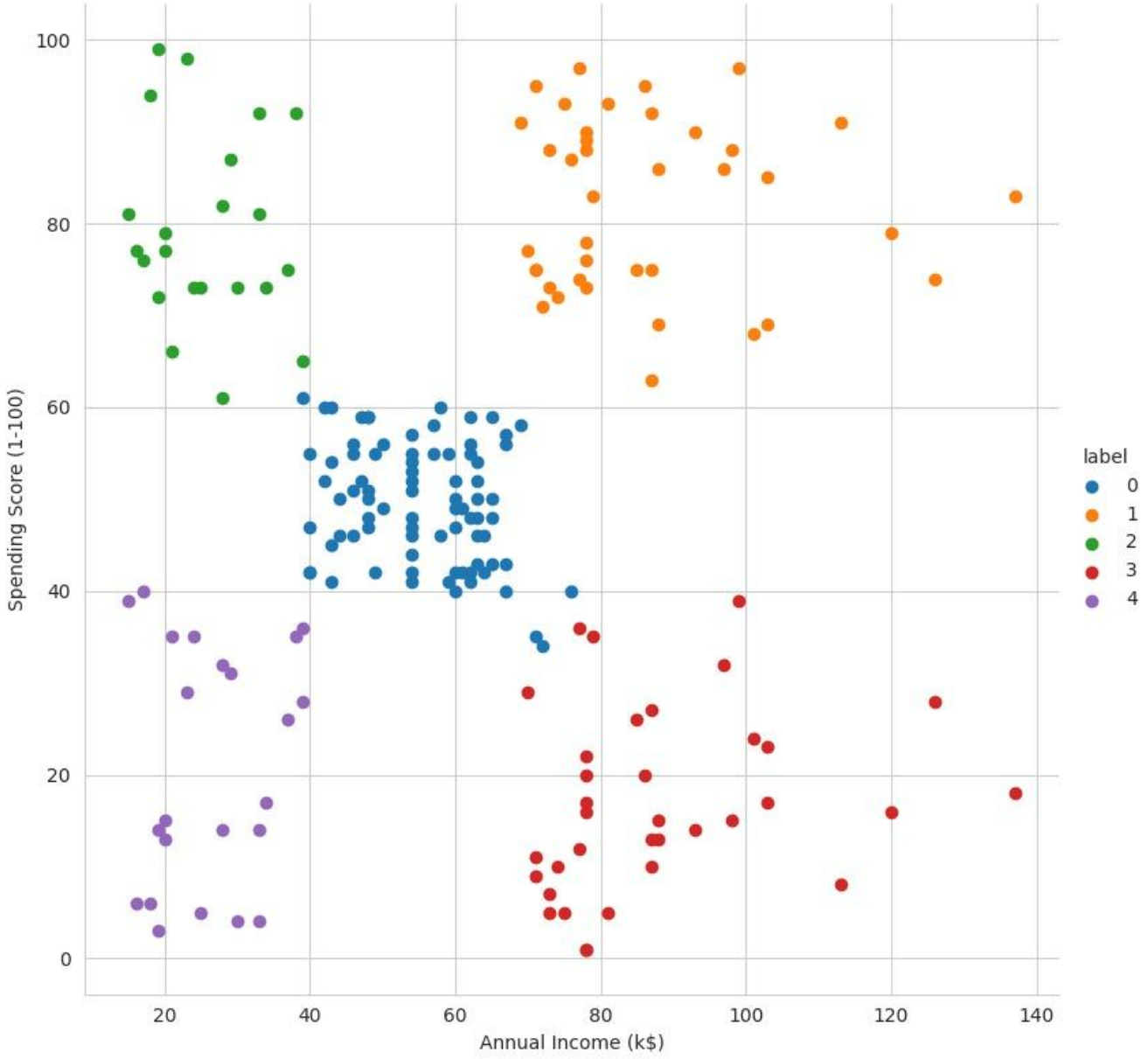
**dtypes: int64(4), object(1)**

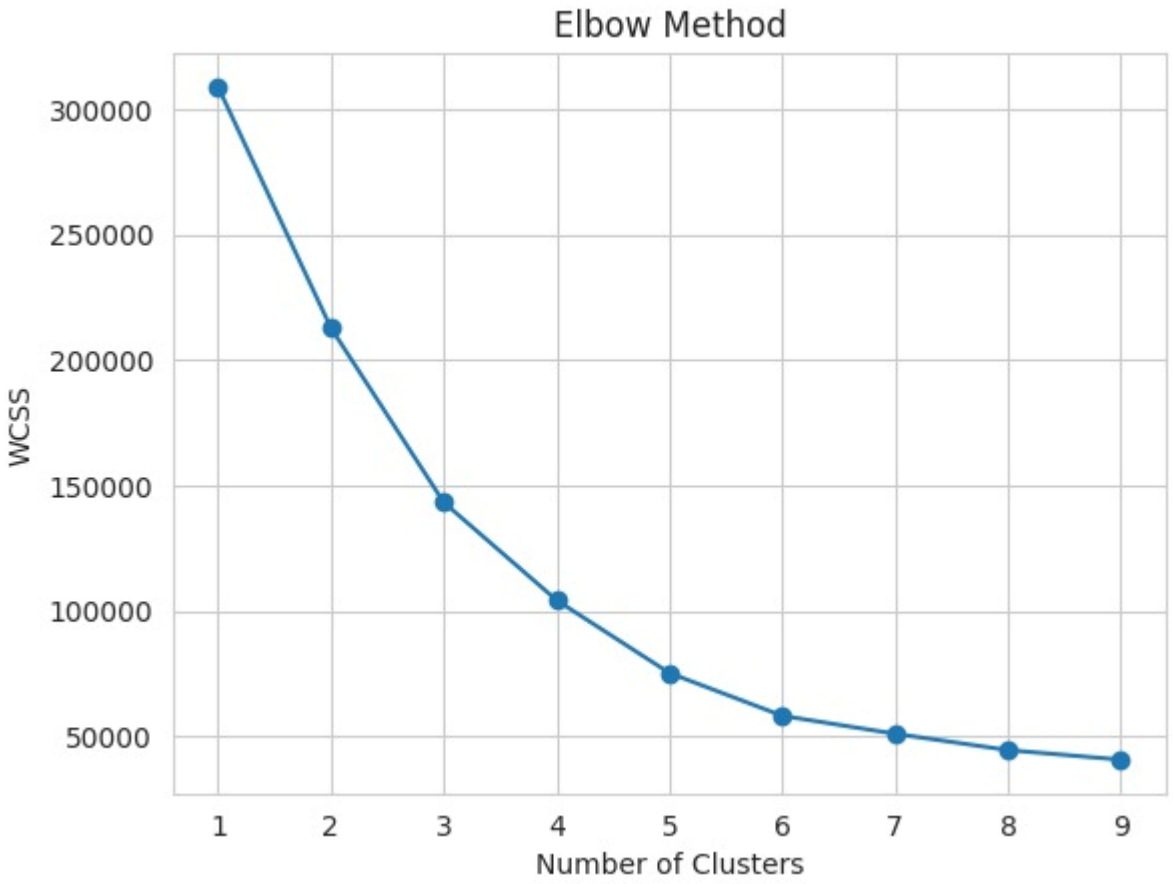
**memory usage: 7.9+ KB**

| | CustomerID | Gender | Age | Annual Income (k$) | Spending Score (1-100) |
|---|---|---|---|---|---|
| 0 | 1 | Male | 19 | 15 | 39 |
| 1 | 2 | Male | 21 | 15 | 81 |
| 2 | 3 | Female | 20 | 16 | 6 |
| 3 | 4 | Female | 23 | 16 | 77 |
| 4 | 5 | Female | 31 | 17 | 40 |




| | Annual Income (k$) | Spending Score (1-100) | label |
|---|---|---|---|
| 0 | 15 | 39 | 4 |
| 1 | 15 | 81 | 2 |
| 2 | 16 | 6 | 4 |
| 3 | 16 | 77 | 2 |
| 4 | 17 | 40 | 4 |


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4
0
20
40
60
80
100
20
40
60
80
100
120
140




**RESULT:**

**Thus, the K-means clustering algorithm was successfully implemented on the given dataset, and the data points were grouped into clusters based on their similarities.**